

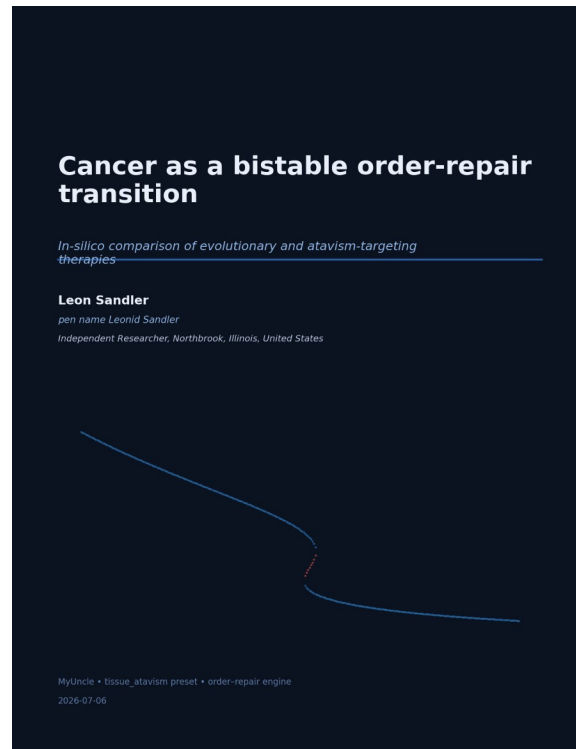
Cancer as a bistable order-repair transition

In-silico comparison of evolutionary and atavism-targeting therapies

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Bistable order–repair motif.

Abstract

We present a theoretical-oncology framework in which cancer initiation and therapy response are governed by a bistable order–repair transition. Our principal innovation is to couple evolutionary (sensitive/resistant) therapy dynamics to a single bistable order parameter I representing multicellular information integrity: healthy tissue is a directionally ordered, cooperative medium eroded by a state-dependent stress flux and restored by repair, so that the mean-field reduction $dI/dt = \mu(I)(1-I) - \rho I$ possesses a low-order attractor identified with the atavistic (quasi-unicellular) phenotype. Across a paired cohort of $n=500$ virtual mice, survival under DNA-damaging protocols increases monotonically from maximum-tolerated dose toward evolutionary-steering plus suppression of stress-induced mutagenesis (A2+A3 Adaptive + SIM-block: median 168 d vs 74 d for MTD), while only protocols targeting recently-evolved vulnerabilities — adaptive immunity and re-differentiation — yield long-term survivors (A1+A4 Immuno + Diff: 33% alive at day 400). A Latin-hypercube global sensitivity analysis ($\pm 30\%$ on nine parameters) preserves the drug-arm ordering in 100% of samples and the young-target survivor advantage in 94%. The model provides a single dynamical framework consistent with the atavism hypothesis's therapeutic corollary — target the

weakness, not the strength — deriving it from dynamical structure rather than tuned assumptions. It is hypothesis-generating rather than confirmatory.

1. Introduction

The atavism (or 'cancer-first') model holds that malignant transformation is a stress-triggered reversion to an ancient, evolutionarily conserved unicellular survival program rather than a purely stochastic accumulation of mutations (Davies & Lineweaver 2011; Lineweaver et al. 2021). Phylostratigraphic analyses report that tumours over-express evolutionarily older genes (Trigos et al. 2017) and that part of cancer's elevated mutation rate is self-inflicted, mediated by homologues of the bacterial SOS / stress-induced-mutagenesis response (Cisneros et al. 2017). If cancer is a reversion, therapy logic inverts: maximally stressing the tumour with DNA-damaging agents both selects for resistance and provokes the ancient mutation engine that generates it (Lineweaver, Davies & Vincent 2014).

1.1 Novelty relative to existing models

Sensitive/resistant competition, adaptive therapy, fitness costs and evolutionary dynamics are individually well established (Gatenby et al. 2009; Zhang et al. 2017). Our contribution is not another resistance model but the coupling of that evolutionary layer to a bistable order–repair variable representing multicellular information integrity. This single addition lets one framework generate, and relate, four therapeutic logics — evolutionary steering, evolvability suppression, differentiation, and immune targeting of young dependencies — and explains why differentiation therapy is bounded by hysteresis and why DNA-stressing therapy is self-defeating, neither of which follows from resistance dynamics alone.

2. Model

2.1 Order–repair engine

Tissue cooperation is summarised by an order parameter $I \in [0,1]$ (high I = cooperative multicellular state; low I = atavistic reversion). Its dynamics follow the mean-field reduction $dI/dt = \mu(I)(1-I) - \rho I$ with self-amplifying repair $\mu(I) = \mu_0(1 + \beta I^h / (K^h + I^h))$ and erosion flux $\rho = \rho_0 + \rho_{\text{str}} \cdot D + \rho_{\text{bur}} \cdot N$ rising with drug DNA-stress D and tumour burden N . The system is bistable (Fig. 1B) and the reversion hysteretic (Fig. 1C): once a large tumour settles into the low-order basin, removing the stress does not restore order — the basis for the limited reach of re-differentiation therapy.

Theory layer: MyUncle order-repair engine on the cancer atavism map

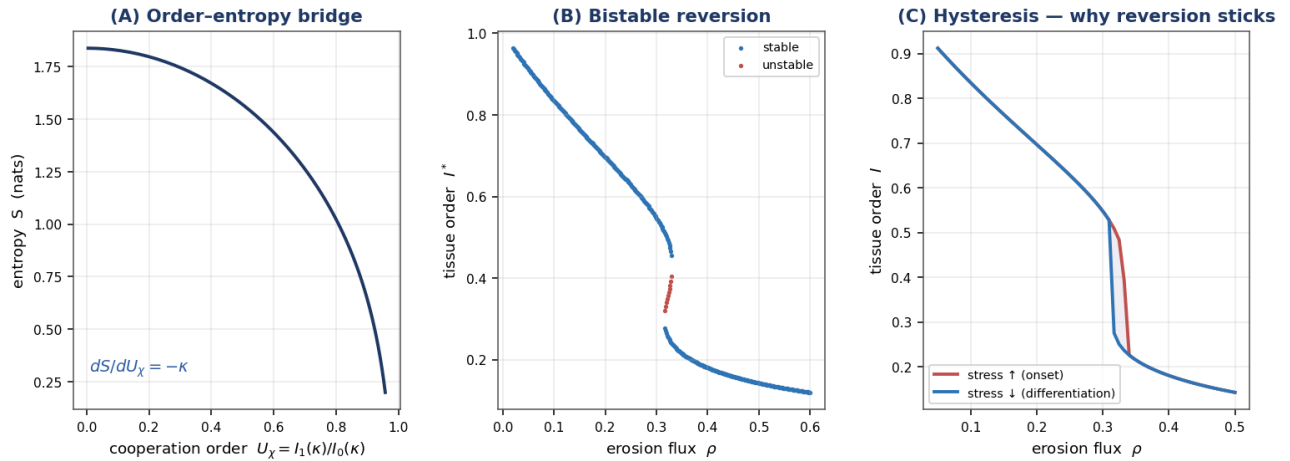


Fig. 1. Theory layer. (A) von Mises order–entropy bridge, $dS/dU_X = -\kappa$. (B) Bistable fixed points $I^*(\rho)$. (C) Hysteresis loop under a stress up/down sweep.

2.2 Evolutionary layer and therapies

Tumour cells partition into drug-sensitive (S) and resistant (R) compartments competing for a shared carrying capacity; resistance carries a fitness cost. Sensitive cells acquire resistance at a baseline rate plus a stress-induced term proportional to dose (the ancient mutation engine), switched off by SIM-inhibitor protocols. Cytotoxic dose kills S; immune clearance targets a recently-evolved dependency and kills S and R alike but erodes with burden (immune escape); re-differentiation adds a flux restoring I. Protocols: MTD; A1 immunotherapy; A2 adaptive dosing (maintaining a high sub-lethal burden so resistant cells are space-limited); A3 MTD + SIM-block; A4 re-differentiation; and combinations A2+A3 and A1+A4.

2.3 Parameter provenance

Parameters are order-of-magnitude phenomenological values chosen to place the system in the biologically relevant regime; sources and rationale are given in Table 1, and robustness is established in Section 3.3 rather than claimed from any single calibration.

Parameter	Value / range	Rationale (source)
g_base	0.05-0.18 /day	Net proliferation; spans reported murine tumour doubling times of days-weeks (phenomenological range).
cost	0.20-0.50	Fitness cost of full resistance; a cost of resistance is documented in vitro and in mice (Bacevic et al. 2017). Swept in Fig. 2D.
phi0	1e-4-5e-3	Pre-existing resistant fraction; consistent with rare standing resistant clones assumed in adaptive-therapy models (Gatenby et al. 2009).

kill_cyto	0.28 /day	Cytotoxic kill of sensitive cells at max dose; phenomenological, chosen so MTD gives an initial response then relapse.
kill_imm / k_esc / imm_floor	0.17 / 0.18 / 0.12	Immune clearance, burden-driven escape, residual surveillance; produce the bimodal hot/cold response seen clinically.
mu0 / mu_sim	8e-5 / 4e-3 /day	Baseline vs stress-induced S->R acquisition; the stress term encodes SOS-homologue mutagenesis (Cisneros et al. 2017).
rho0 / rho_str / rho_bur	0.05 / 0.25 / 0.32	Order-erosion flux from baseline, drug DNA-stress, and burden; sets the bistable/hysteretic regime (Fig. 1).
diff_flux / rev_diff	0.12 / 0.04	Re-differentiation drive and R->S reversion; phenomenological analogue of differentiation therapy (e.g. retinoic acid in acute promyelocytic leukaemia).

Table 1. Parameter provenance.

3. Results

3.1 Two families of outcomes

Among cytotoxic protocols median survival rises monotonically MTD < MTD+SIM-block < adaptive < adaptive+SIM-block, so containment plus evolvability-suppression outperforms maximal cytoreduction. Yet the only protocols yielding long-term survivors attack cancer's young vulnerabilities: immunotherapy clears immunogenic tumours outright (bimodal), and immunotherapy with re-differentiation leaves the largest surviving fraction. Re-differentiation alone delays death longest but rarely holds, consistent with the order–repair hysteresis.

In-silico horse race: order-repair engine + evolutionary tumor dynamics

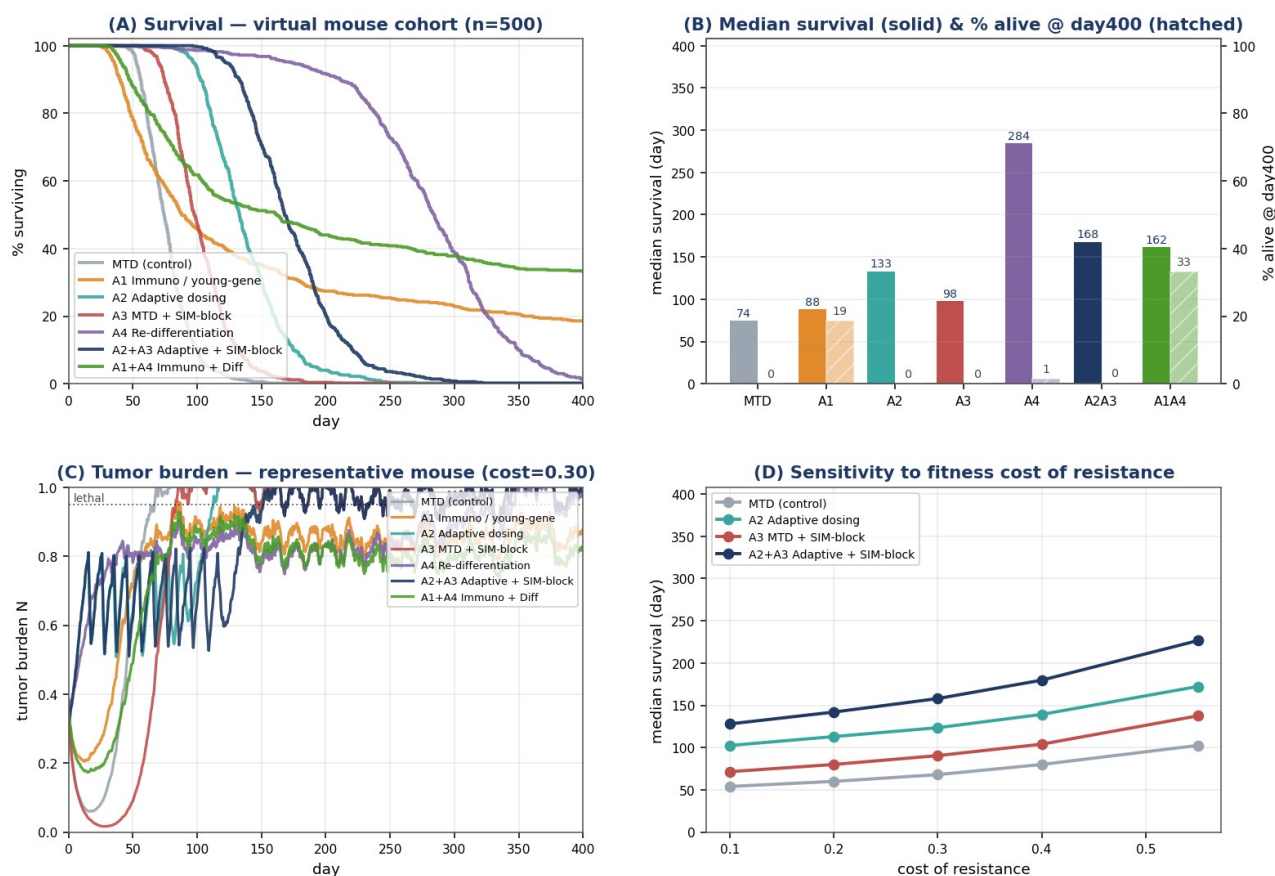


Fig. 2. Horse race. (A) Survival. (B) Median survival and % alive at day 400. (C) Representative-mouse burden trajectories. (D) Sensitivity to fitness cost of resistance.

3.2 Face validation against reported findings

The model is not calibrated to data, but its qualitative behaviour reproduces independently reported results, a minimal consistency check. (i) MTD produces an initial response then resistant relapse, and adaptive dose-modulation prolongs control — as in preclinical mouse studies and the metastatic castrate-resistant prostate-cancer trial of adaptive therapy (Gatenby et al. 2009; Zhang et al. 2017). (ii) The advantage of adaptive dosing depends on a fitness cost of resistance (Fig. 2D), a cost demonstrated in vitro and in mice with a CDK inhibitor (Bacevic et al. 2017). (iii) Stress-induced mutagenesis accelerating resistance reflects the SOS-homologue mutation signature reported in cancer (Cisneros et al. 2017). (iv) The immune arm's bimodal hot/cold outcome mirrors clinical immunotherapy heterogeneity, and the superiority of young-dependency targeting matches the atavistic 'target the weakness' prediction (Lineweaver, Davies & Vincent 2014). Reproducing these without fitting to any of them is weak but non-trivial evidence that the dynamical structure is biologically sensible.

3.3 Global sensitivity

A Latin-hypercube sample (n=120) varied nine mechanistic parameters simultaneously by $\pm 30\%$. The drug-arm ordering (MTD $\leq$ SIM-block $\leq$ adaptive $\leq$ adaptive+SIM-block) is preserved in 100% of samples, and the young-target family (immunotherapy $\pm$ differentiation) provides the most long-term survivors in 94%. A one-at-a-time tornado (Fig. 3B) shows the adaptive+SIM-block survival advantage over MTD is most sensitive to the stress-induced mutagenesis rate and the burden-driven erosion flux, and never reverses sign within the range tested. The qualitative conclusions are therefore properties of the model structure, not of a single parameter choice.

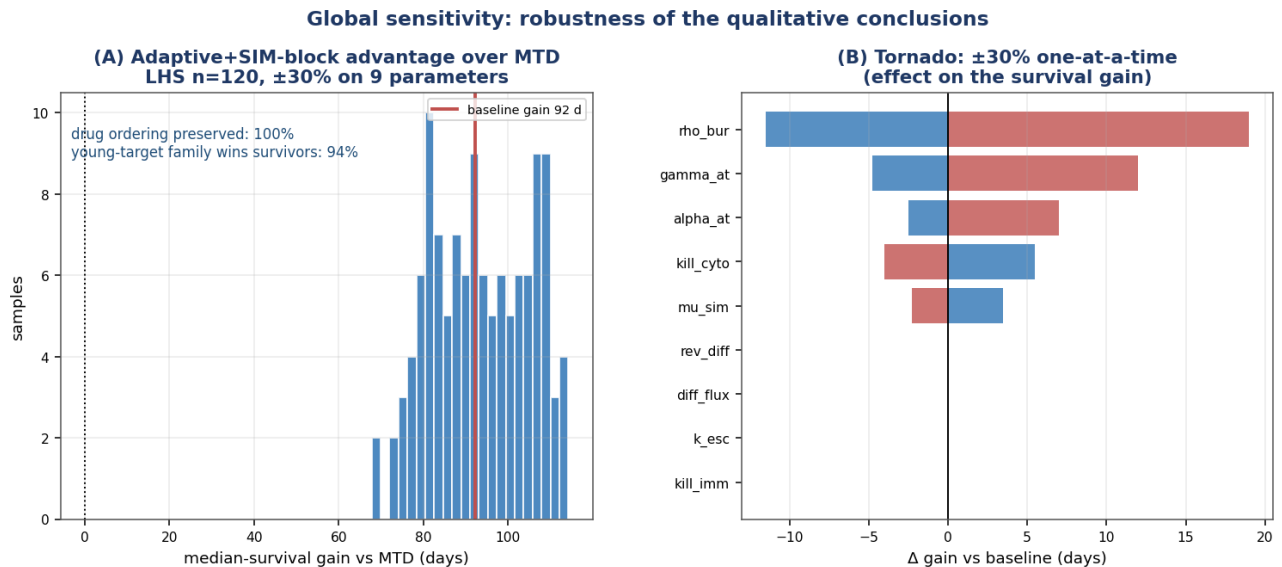


Fig. 3. Global sensitivity. (A) Distribution of the adaptive+SIM-block survival advantage over MTD across Latin-hypercube samples. (B) One-at-a-time $\pm 30\%$ tornado.

3.4 Protocol summary

Protocol	Median survival (d)	% alive @400	% durable regression	Mean dose
MTD (control)	74	0.0	0.0	400
A1 Immuno / young-gene	88	18.6	0.2	0
A2 Adaptive dosing	133	0.0	0.0	312
A3 MTD + SIM-block	98	0.0	0.0	400
A4 Re-differentiation	284	1.4	0.0	40
A2+A3 Adaptive + SIM-block	168	0.2	0.0	282
A1+A4 Immuno + Diff	162	33.2	0.8	0

Table 2. Outcomes by protocol.

4. Discussion

4.1 Relation to other frameworks

The somatic mutation theory treats cancer as accumulated driver mutations under selection; our model is compatible with it but adds a macroscopic order variable that the SMT lacks. Compared with cancer-stem-cell and tissue-organisation-field views, the order parameter I plays a role analogous to a coarse-grained 'degree of tissue organisation', while our therapy layer inherits the ecological/evolutionary logic of adaptive therapy (Gatenby et al. 2009). The novel content is the bistability and hysteresis of I , which supply mechanisms — self-defeating DNA-stress therapy and hysteresis-bounded differentiation — absent from purely evolutionary models.

4.2 Proposed protocol and test

The model points to a layered protocol: an adaptive + SIM-block cytotoxic backbone to contain bulk disease, hold evolvability down and keep cells re-sensitisable, combined with immunotherapy and re-differentiation to clear the contained, more-immunogenic residual.

4.3 Predictions (falsifiable)

P1. In a syngeneic mouse model (e.g. 4T1, B16), adaptive dosing plus a translesion-synthesis / SIM inhibitor extends time-to-resistance beyond adaptive dosing alone and beyond MTD. P2. Barcoded clonal diversity rises fastest under MTD and slowest under SIM-block arms. P3. Re-differentiation efficacy declines with initial tumour burden (a hysteresis threshold), not merely with growth rate. P4. Adaptive therapy's benefit scales with the measured fitness cost of resistance across cell lines. Any of these failing would falsify the corresponding model mechanism. As a direct test of the underlying atavism premise, the prediction that tumours over-express evolutionarily older (unicellular-age) genes and down-regulate younger (multicellular-age) genes (Domazet-Loso & Tautz 2010; Trigos et al. 2017) can be evaluated quantitatively on public tumour/normal expression data; an analysis pipeline for this is provided (Section 5).

4.4 Limitations

We stress that the therapeutic rankings are model-generated hypotheses to be tested (Section 4.3), not evidence that any protocol is clinically superior; the model derives which predictions should hold and under what conditions, not that they are true. This is an in-silico model — the ranking follows from encoded mechanisms and is not empirical data. Parameters are phenomenological (Table 1) and, although the conclusions are robust across the ranges tested (Section 3.3), they are not fitted to a specific tumour. The two-type S/R approximation omits a continuous resistance spectrum and spatial structure. The atavism framework itself remains contested, and phylostratigraphic inference is sensitive to overlapping gene-age categories, so downstream genomic claims should be guarded against overlap-induced significance inflation.

5. Code and data availability

The differential-equation and stochastic-cohort simulations were implemented in a custom Python framework. All code, parameter files, and scripts to regenerate every figure and table are openly archived (Zenodo DOI to be assigned on deposit; mirror on GitHub) to allow full reproduction. The distribution also includes a phylostratigraphy analysis pipeline that assigns gene ages, tests the unicellular-versus-multicellular expression shift between tumour and normal samples, and controls for gene-gene correlation via a sample-label permutation null to avoid overlap-induced significance

inflation; it accepts standard expression matrices (e.g. TCGA/GTEX) for direct empirical evaluation of the atavism premise.

Declarations

Funding: This work received no external funding.

Conflicts of interest: The author declares no competing interests.

Ethics: This study is entirely computational and involved no human participants or animal subjects.

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